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METHOD

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Special Collection:

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Key Points:

- We propose an AI-facilitated near real-time monitoring framework that has been deployed at TexNet using a recently developed deep learning (DL) picker
- Due to the fast-increasing seismicity in West Texas, this AI-facilitated framework significantly relieves the workload of TexNet analysts
- The mean absolute error (MAE) of automatic magnitude estimation for the magnitude-above-two earthquakes is smaller than 0.15 in West Texas

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Near Real-Time Earthquake Monitoring in Texas Using the Highly Precise Deep Learning Phase Picker

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Abstract Artificial intelligence (AI) seismology has witnessed enormous success in a variety of fields, especially in earthquake detection and *P* and *S*-wave arrival picking. It has become widely accepted that DL techniques greatly help routine seismic monitoring by enabling more accurate picking than traditional pickers like STA/LTA. However, a completely automatic AI-driven earthquake monitoring framework has not been reported due to the concerns of potential false positives using DL pickers. Here, we propose a novel AI-facilitated near real-time monitoring framework using a recently developed deep learning (DL) picker (EQCCT) that has been deployed in the Texas seismological network (TexNet). For the West Texas area, TexNet's seismic monitoring relies on the EQCCT picker to report earthquake events. For earthquakes with a magnitude above two, the picks are further validated by analysts to output the final TexNet catalog. Due to the fast-increasing seismicity caused by continuing oil&gas production in West Texas, this AI-facilitated framework significantly relieves the workload of TexNet analysts. We show the mean absolute error (MAE) of automatic magnitude estimation for the magnitude-above-two earthquakes is smaller than 0.15 in West Texas and MAEs of hypocenter locations within 2.6 km in both distance and depth estimates. This research provides more evidence that DL pickers can play a fundamental role in daily earthquake monitoring.

Plain Language Summary AI has been a significant part of almost every aspect of our life. However, it is not widely accepted that AI can be a game changer in geoscience. In this work, we provide a compelling example that AI can significantly lower the overload of earthquake analysts in everyday work and boost our earthquake detectability, thereby enhancing our understanding of earthquake statistics, physics, and potential nucleation mechanisms. In a nutshell, AI has already been demonstrated to be successful in earthquake analysis but never in real-time monitoring, which requires a high success rate and zero tolerance to large-earthquake detection errors. Here, we show that AI-assisted earthquake monitoring workflow can be almost 100% accurate for moderate-to-large earthquakes.

1. Introduction

Texas exhibits the most frequent moderate-size earthquakes (e.g., $M > 2$) in the United States because of the relentless industrial activities, including injection and extraction of fluid in several major hydrocarbon accumulated basins (Doser et al., 1992; Frohlich & Davis, 2003; Frohlich et al., 2016; Skoumal et al., 2020). While in the past decade, a majority of the earthquake events have been from the Delaware basin, which is mostly unpopulated areas, there is a trend that the seismicity is merging in the populated areas like the Midland basin in the past 3 years, including the 2022 Magnitude 5.4 Range Hill event. Specifically, the number of $M > 2$ events is larger than 200 per month, and $M > 3.5$ events happen almost every week statewide. Therefore, real-time seismic monitoring in Texas has become particularly demanding and is crucial for earthquake hazard mitigation and management (Amato et al., 2006; Bansal et al., 2021; Hanka et al., 2010; Lee et al., 2014; Zhang et al., 2014).

In general, real-time seismic monitoring systems need to provide detailed information to let the stakeholders be aware of the existence and the evolution trend of the seismicity. For large destructive earthquakes, the mission of a real-time monitoring system is to provide accurate source information, including (epicenter location, depth, origin time, magnitude, and focal mechanisms) with minimal time delay. The short acceptable time delay of real-time seismic monitoring systems set up a high standard for the accuracy of automatic earthquake detectors and phase pickers. Traditionally, the Texas seismological network (TexNet) (Savvaidis et al., 2019) is based on an enhanced STA/LTA picker (Lomax et al., 2012), which is reasonably high performance in detecting most large-size

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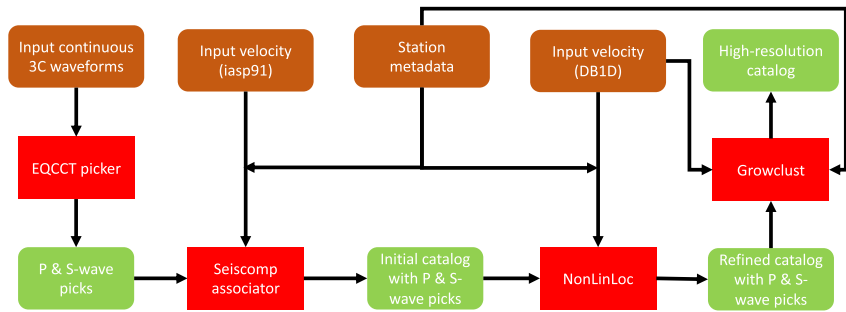


Figure 1. Workflow for creating a high-resolution catalog using the deep learning picker. In the framework, there is one intermediate catalog, which is directly from the SeisComP software and iasp91 velocity model. The NonLinLoc with a regional 1D velocity model (DB1D) is used to refine the catalog to correct the depth errors. Here, we only consider the generation of the hypocenter locations of detected earthquake events, while the magnitude estimation is skipped from this workflow and is based on the method developed in Kavoura et al. (2020).

earthquakes and a huge number of moderate-size events. In the meantime, despite the high recall rate in large earthquake events, the *P* and *S* wave arrival times cannot be picked with high accuracy due to the complex waveform features and high noise labels in the field. As a result, after large earthquakes ($M > 3.5$), significant intervention by TexNet analysts is required to adjust the arrival times to outsource a high-quality hypocenter location and estimate the magnitude of the earthquake. In addition to the moderate-size earthquakes, there are a large number of false alarms within the catalog using the traditional STA/LTA picker.

Considering the fast-increasing seismicity in all magnitude ranges due to continuing oil&gas production, manual adjustment of the arrival pickers becomes prohibitively labor-intensive and should better be minimized. Deep learning phase pickers have been widely studied in the past decade and have undergone many case verifications from around the world (Saad et al., 2023; Woollam et al., 2022). Many studies have shown that deep learning phase pickers can be far more accurate than traditional STA/LTA pickers (Lomax et al., 2012; Trnkoczy, 2009) or more advanced template matching-based pickers (Mu et al., 2017; Yoon et al., 2015). Notably, Zhu et al. (2023) developed a cloud-based automatic earthquake detection and location workflow using PhaseNet. However, there is a common issue in most of the deep learning pickers, that is, despite the high detection recall, there might still be high picking errors (e.g., early or delayed picks) within the picked *P* and *S*-wave phases. As for the rapid response to large-magnitude events, analysts' adjustments are still required to ensure high location accuracy. For small and moderate-size events, a sophisticated workflow that includes both location and relocation steps is used for correcting inaccurate arrival picks. Here, we demonstrate the high picking accuracy of a newly developed deep learning picker, EQCCT (Saad et al., 2023), in two scenarios, that is, moderate-to-large events ($M > 2$) (hereafter referred to as M2 events) and other smaller events. For the moderate-to-large events, we demonstrate that location and magnitude estimation arising from deep learning pickers are sufficiently accurate for rapid response purposes. For the smaller events, we demonstrate that through an integrated workflow, we can output a highly complete catalog in near real time.

2. Method

2.1. EQCCT - The Latest Deep Learning Phase Picker

The transformer is a powerful tool for extracting important information from input data. EQCCT, based on a compact Convolutional Transformer (CCT), can identify the locations of *P* and *S*-wave arrival times within the input data. EQCCT uses a sample-based picking approach, assigning an output label to each input sample. It employs a triangle function centered around *P* and *S*-wave arrival times as the target label. EQCCT is a newly developed phase picker first published in Saad et al. (2023). It is based on the latest deep learning architecture, that is, compact convolutional transformer (CCT), which combines the benefits of the first-generation convolutional network as used in PhaseNet and the second-generation transformer network as used in EQTransformer and overcomes their individual drawbacks. For example, EQCCT can be more effective in extracting the time-series feature than PhaseNet and can be faster than EQTransformer. In (Saad et al., 2023), many detailed benchmark comparisons with PhaseNet Zhu et al. (2023), and EQTransformer (Mousavi et al., 2020) in terms of picking

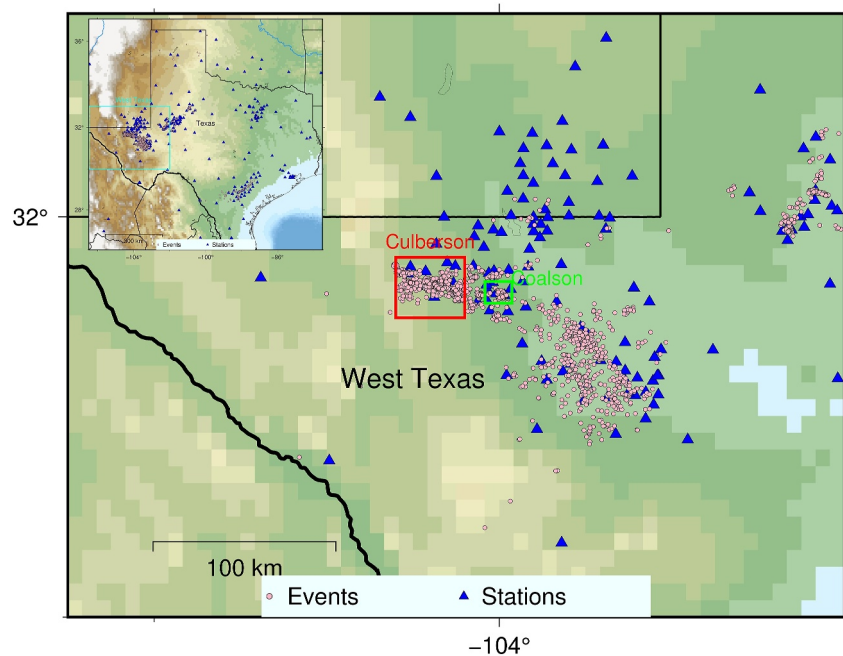


Figure 2. Map of the Texas stations and event distributions. The green frame box indicates the area of interest of the Coalson monitoring example. The red frame box indicates the area of interest of the Culberson monitoring example. The inset shows the event and station distribution of the whole state of Texas. The cyan frame box highlights the area of West Texas studied in this paper.

accuracy on different testing datasets have been conducted. Here, we will focus on its application in near real-time monitoring at TexNet.

EQCCT consists of two independent models, one for *P*-wave picking and the other for *S*-wave. Both models are trained independently using the same dataset. Training involves an augmented version of the STEAD dataset (Mousavi et al., 2019), with EQCCT taking 60 s three-component seismograms as input. Augmentation techniques include shifting, adding a second earthquake in the same 60 s window, adding noise, and corrupting seismograms by dropping one or two components. This augmentation improves the generalization ability of the EQCCT when applied to different datasets.

Convolutional layers extract feature maps from the input data, followed by a patching layer that divides the features into non-overlapping windows. EQCCT also includes transformer blocks to enhance the extracted features. The core layer is the Multi-Head Attention (MHA) network, which assigns higher attention to the important information, particularly the location of arrival times. MHA comprises four self-attention heads, and the output is the combination of these heads. Dropout layers, like Stochastic Depth Dropout (SDD) layers (Loshchilov & Hutter, 2016), are integrated to prevent overfitting.

As a result, EQCCT demonstrates strong picking performance with a small output threshold (e.g., 0.1) and remains reliable when applied to various datasets (Saad et al., 2023). The overlap ratio between neighboring 60 s windows in default is set to 0.3.

2.2. General Workflow for Near Real-Time Monitoring

Following single-station-based earthquake detection, *P* and *S*-wave arrival picking, we design an integrated workflow shown in Figure 1 to locate and relocate the earthquake events. The complete open-source earthquake monitoring framework can be found at <https://github.com/chenyk1990/sc3flow>. After the *P* and *S*-wave picks are obtained using the EQCCT method, we apply the state-of-the-art SeisComP software with the LOCSAT locator and the iasp91 velocity model (Kennett et al., 1995) to associate the *P* and *S*-wave picks and obtain the initial hypocenter locations. The proposed AI detection module is connected to the SeisComp messaging system. It does not listen to seedlink directly in real time, instead it reads from the archive through FDSNWS with a short

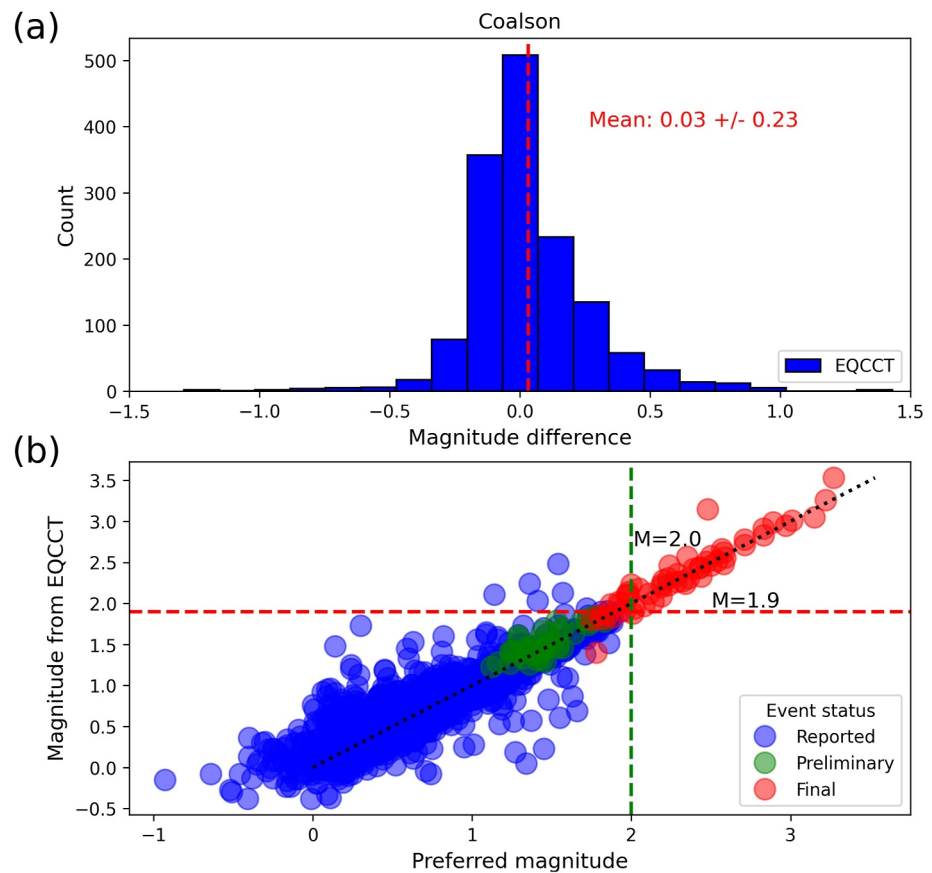


Figure 3. Monitoring test in Coalson area. (a) Magnitude error distribution. (b) Scatter plot of EQCCT magnitude V.S. preferred magnitude. Preferred magnitude denotes the most accurate magnitude decided at TexNet using the most reliable picks, velocity models, and location workflows.

processing latency (currently of 10 min). The tests shown in this paper are all based on SeisComp3, which is no longer used by TexNet but is completely compatible with SeisComp6. The SeisComp6 version of this AI detection workflow has been run by TexNet recently. Based on the examples shown later, we found that the recall values for $M > 2$ earthquakes are almost 100%, indicating significantly reduced arrival picking errors compared with state-of-the-art automatic methods.

The association and location procedures are iterated until the optimal association and hypocenter location are obtained when the P and S -wave traveltime residuals are minimized. To correct the depth errors and refine the hypocenter locations, we further apply the NonLinLoc method (Lomax et al., 2009) with a regional 1D velocity (DB1D) to apply a secondary location on the associated picks from the last step using LOCSAT. To facilitate a fair comparison with the TexNet catalog, we use exactly the same location workflow as the routine TexNet monitoring, including association, location, and refined location, except for different P and S -wave picks, and the same parameters for all individual tasks through the workflow. In previous TexNet daily monitoring, we used an enhanced STA/LTA picker while in the new TexNet daily monitoring, we use the EQCCT as the picker. For moderate-to-large ($M > 2$) earthquakes, the earthquake catalog after the NonLinLoc step is output for manual quality control before being put online in the public TexNet catalog. For small ($M < 2$) earthquakes, the high-resolution catalog after the relocation step is output as the final product. The whole workflow detailed in Figure 1 is completely automatic.

3. Examples

We conduct tests in two regions, Coalson and Culberson. The studied areas are indicated in Figure 2 as the green and red frame boxes. Figure 2 also plots all the TexNet recorded events and station distributions. In the state of

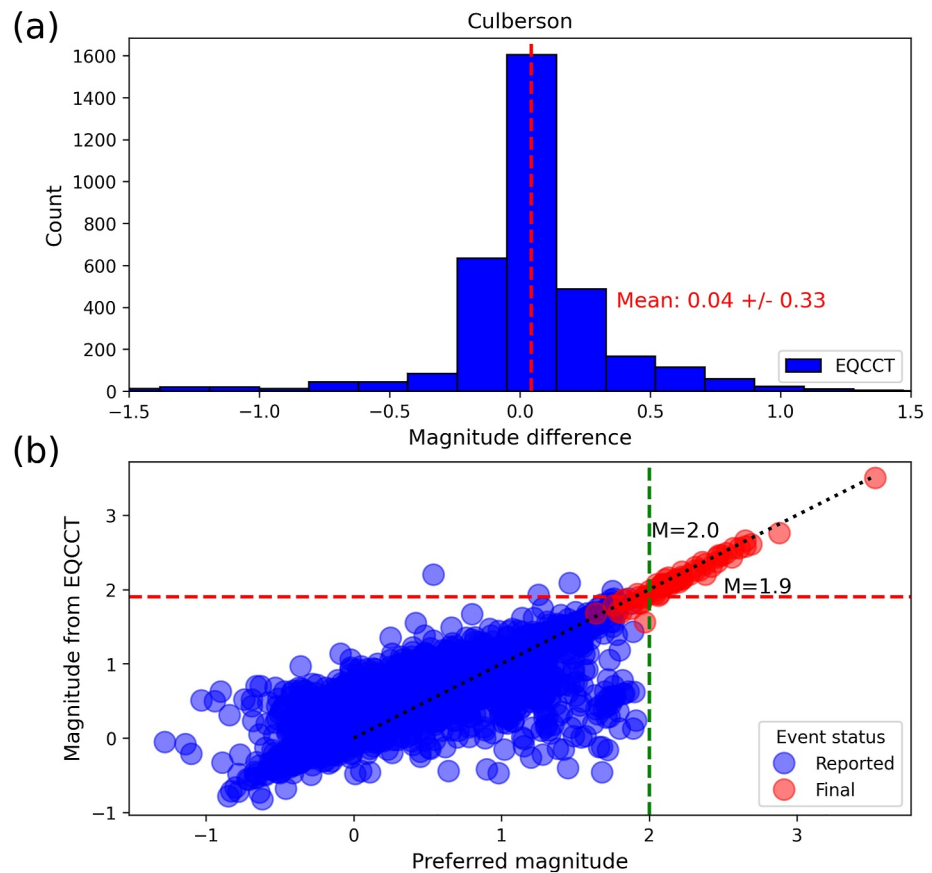


Figure 4. Monitoring test in Culberson area. (a) Magnitude error distribution. (b) Scatter plot of EQCCT magnitude V.S. preferred magnitude. Preferred magnitude denotes the most accurate magnitude decided at TexNet using the most reliable picks, velocity models, and location workflows.

Texas, there are several seismically active basins, here, we only study the Delaware basin region in West Texas, indicated by the cyan frame box in the inset of Figure 2.

The two monitoring examples are shown in Figure 3 for Coalson and 4 for Culberson. We chose the period between 2022/11/28 and 2022/12/31 to study the near-real-time monitoring performance. Here, we mainly study the magnitude estimation accuracy (Kavoura et al., 2020) because the magnitude is the primary factor that we consider regarding whether manual picking is performed on an automatically detected event in the Coalson area (depicted by the green frame box in Figure 2) and the given period, we detect 1,471 events, with a minimum magnitude around -1 Ml and a maximum above 3 Ml. Here, we compare the EQCCT magnitude with the preferred magnitude. The preferred magnitude means the best magnitude estimation for a certain event, either from Texnet analysts' manual picking or from a rigorous location procedure. The magnitude from EQCCT denotes the magnitude output directly from the automatic EQCCT workflow. The magnitude difference distribution is plotted in Figure 3a, where it is clear that the magnitude difference is mostly around the zero coordinate. The mean absolute error of magnitude estimation is 0.03 and the standard deviation is 0.23. The red dashed line indicates the mean magnitude difference (0.03). Figure 3b shows the scatter plots between EQCCT magnitude and preferred magnitude in three cases, that is, reported, preliminary, and final. Reported events are events that are directly output from the EQCCT-based workflow without going through the manual picking process. Preliminary events mean that the events have been manually reviewed by Texnet analysts but have not been finalized. The final events mean the events that have gone through several analysts' checks and have been considered as the best solutions. We can see that both preliminary and final events have very small errors using EQCCT, meaning that only a few changes were made during the manual picking process. Note that $M = 1.9$ is the threshold for manually checking the EQCCT-triggered events. Above magnitude 1.9, events will go through a manual check by TexNet analysts. In this Coalson example, 43 M2 events were observed after manual checks. There is only one $M < 1.9$

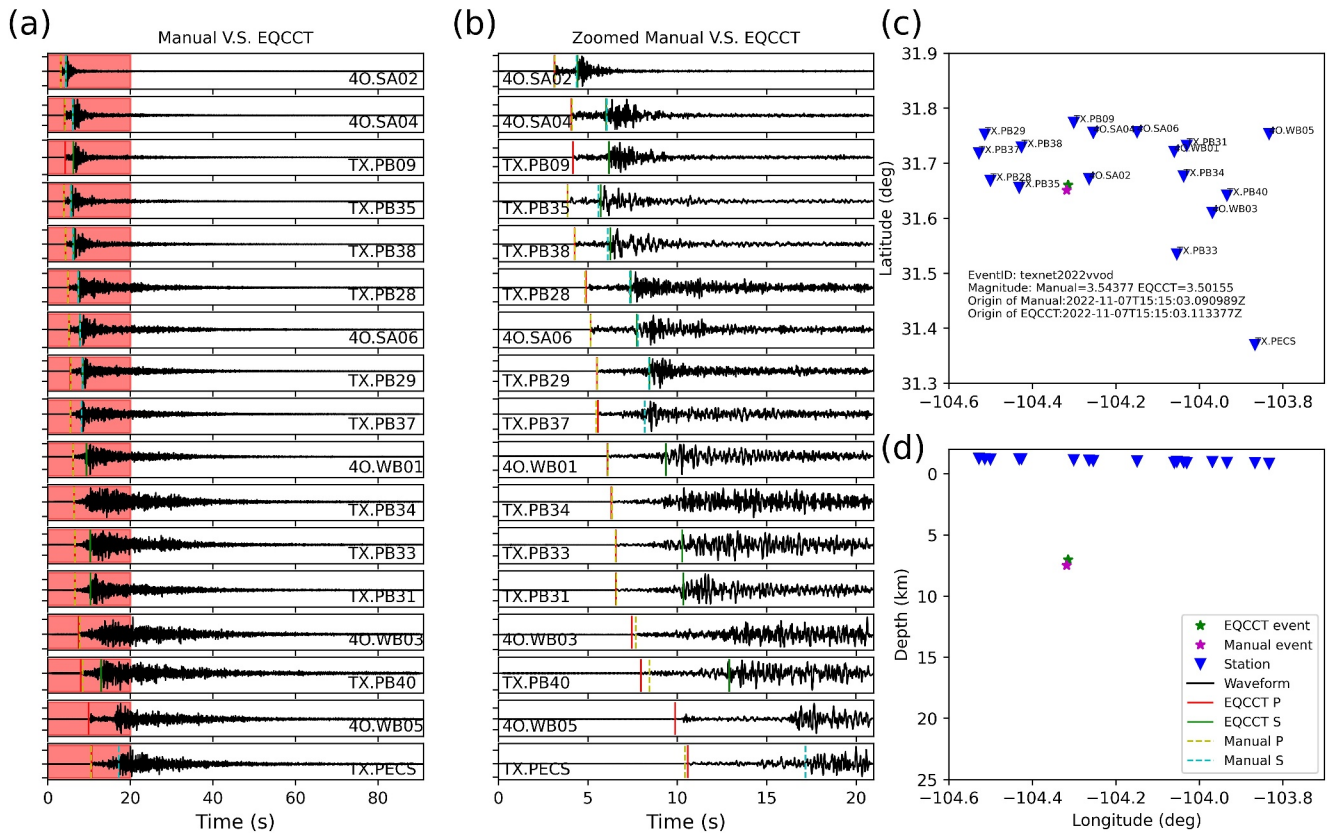


Figure 5. Waveform comparison for a large earthquake ($M = 3.5$). (a) Waveforms in different stations corresponding to the TexNet event texnet2022vvod with picks from EQCCT and manual picks on top of the waveforms (vertical lines). (b) Zoomed comparison (the first 20 s). (c) Station distribution and event location comparison on the map view. (d) Station distribution and event location comparison on the depth slice. The near real-time EQCCT monitoring result is almost the same as the manually picked result in terms of magnitude (error <0.05), origin times (error <0.02 s), hypocenters (error <0.5 km for longitude, latitude, and depth).

event from EQCCT ($M_{EQCCT} = 1.88$) with a final magnitude above 2 ($M_{manual} = 2.01$), meaning that the recall of M2 events is 97.7%. It is worth noting that there is a human factor in the quality control process as manual picks are conducted by multiple analysts, so the manual results are not strictly the ground truth but are relatively robust.

Then, we test the near real-time monitoring performance in Culberson County. With the same monitoring period and the geographic area of interest depicted by the red frame box in Figure 2, we detect 3,350 events, with a minimum magnitude below -1 Ml and a maximum above 3.5 Ml. The magnitude difference distribution is plotted in Figure 4a. The mean absolute error of magnitude estimation is 0.04, and the standard deviation is 0.33. The red dashed line indicates the mean magnitude difference (0.04). Figure 4b shows the scatter plots between EQCCT magnitude and preferred magnitude in two cases, that is, reported and final. In this example, preliminary events have all been marked as final events. Similarly, we can see that all final events have very small errors using EQCCT, meaning that EQCCT picks are highly accurate. Specifically, in this Culberson example, there are 45 M2 events after a manual check. There are no $M < 1.9$ events from EQCCT with a final magnitude above 2, meaning that the recall of M2 events is 100%.

We then compare the near real-time picking results using EQCCT on earthquakes in two scenarios: large earthquakes and small earthquakes. Figure 5 shows the comparison of picking and location results between TexNet analysts (manual) and EQCCT. Figure 5a shows the 90 s waveforms from 17 stations for the M3.5 event (texnet2022vvod) on the TexNet public catalog. We plot the manual (dashed lines) and EQCCT (solid lines) picks on top of the waveforms as the vertical lines. In general, the manual and EQCCT picks (P and S) almost overlap each other. For a better and more detailed view, we zoom the waveforms, highlighted by the red rectangles, in Figure 5b, where we can see more clearly the overlapping or mismatch between manual and EQCCT picks. Most picks have very small picking errors (e.g., <0.05 s), while only a few have relatively larger and more observable errors. The largest picking errors appear on the TX.PB40 station (third last from bottom to top), possibly because

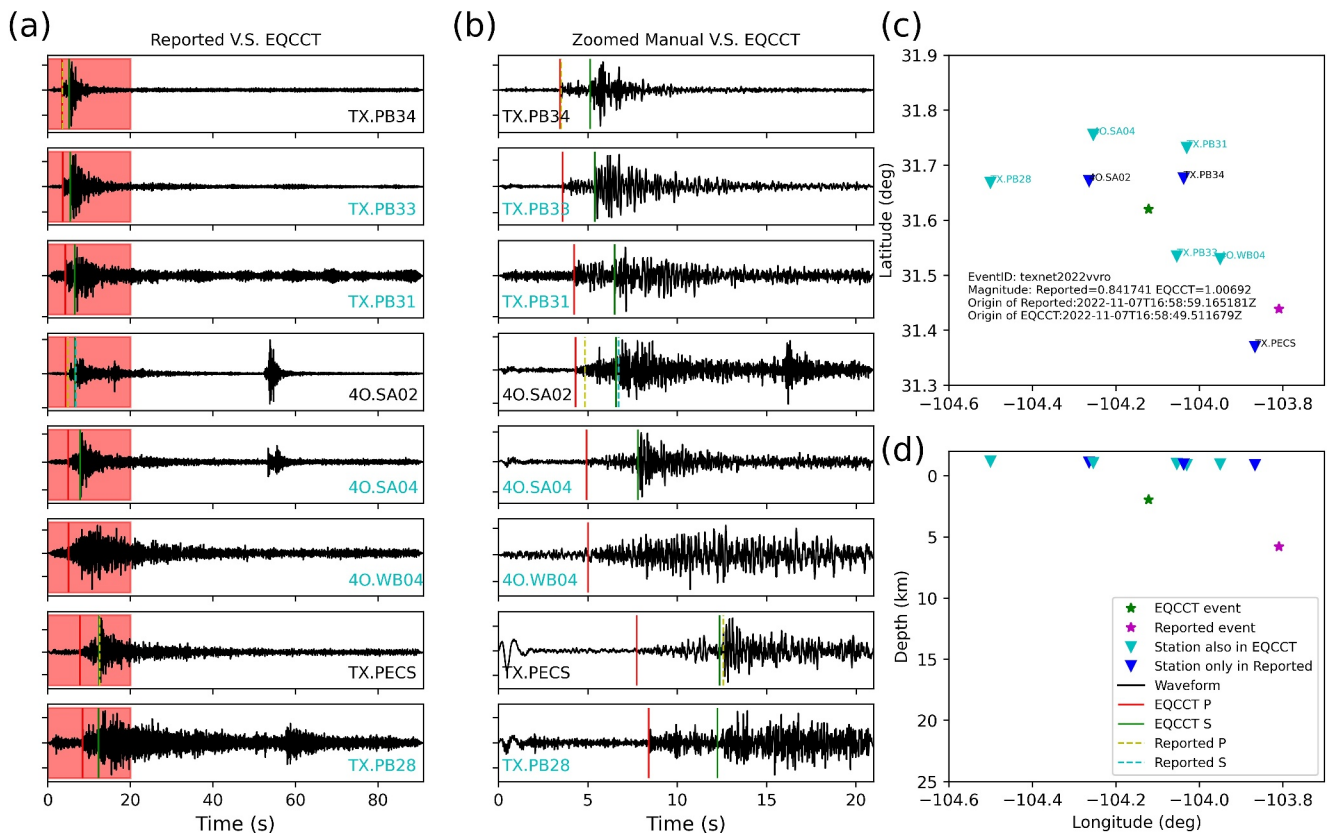


Figure 6. Waveform comparison for a small earthquake ($M < 1$). (a) Waveforms in different stations corresponding to the TexNet event texnet2022vvro with picks from EQCCT and reported picks on top of the waveforms (vertical lines). (b) Zoomed comparison (the first 20 s). (c) Station distribution and event location comparison on the map view. (d) Station distribution and event location comparison on the depth slice. The reported picks here indicate using a traditional, well-optimized picker (Lomax et al., 2012). In this case, the traditional picker only detects the event in three stations highlighted by blue while the EQCCT detects in eight stations. The arrivals are obviously more accurate using EQCCT.

of its relatively higher noise level. For the most part, EQCCT picks are highly accurate compared with manual picks, which makes the hypocenter locations of the two scenarios very close (< 0.1 km in latitude, longitude, and depth). The location comparisons and station distributions are shown in Figures 5c and 5d. The magnitude error is only 0.04, and the origin time error is only 0.02 s. This example demonstrates that EQCCT is sufficiently accurate in near real-time monitoring for large earthquake events (e.g., $M > 3.5$ events, which requires event characterization almost immediately after the occurrence).

We then show another small magnitude ($M < 1.0$) event (texnet2022vvro) in Figure 6. This event is detected by TexNet traditional automatic picker, which is an optimized STA/LTA method (Lomax et al., 2012), and is labeled as a reported event. Because it has a very small magnitude, TexNet analysts do not manually pick the events. Here, we only compare the EQCCT picks with the reported picks (from the traditional picker) and the EQCCT location with the reported location (from the automatic association and location workflow based on the traditional picker). Similarly, Figure 6a plots the 90 s waveforms and the EQCCT (solid vertical lines) and reported picks (dashed vertical lines). Figure 6b shows the zoomed comparison between the picks. Notably, the traditional picker only detects the event from three stations, while EQCCT detects the event from eight stations. More importantly, EQCCT almost picks P and S arrivals from all the stations. From observation, it is obvious that EQCCT picks are more accurate than the picks from the traditional picker. The stations that are not triggered by the traditional picker are highlighted by the cyan labels on the right or left part of each waveform in (a) or (b). The corresponding stations are also plotted in 6(c) and 6(d) as the cyan inverted triangles. The blue inverted triangles show the three stations that are triggered by the traditional picker. As a result, the reported location much deviates from the EQCCT location. From a comprehensive viewpoint, the EQCCT automatic location for this small earthquake

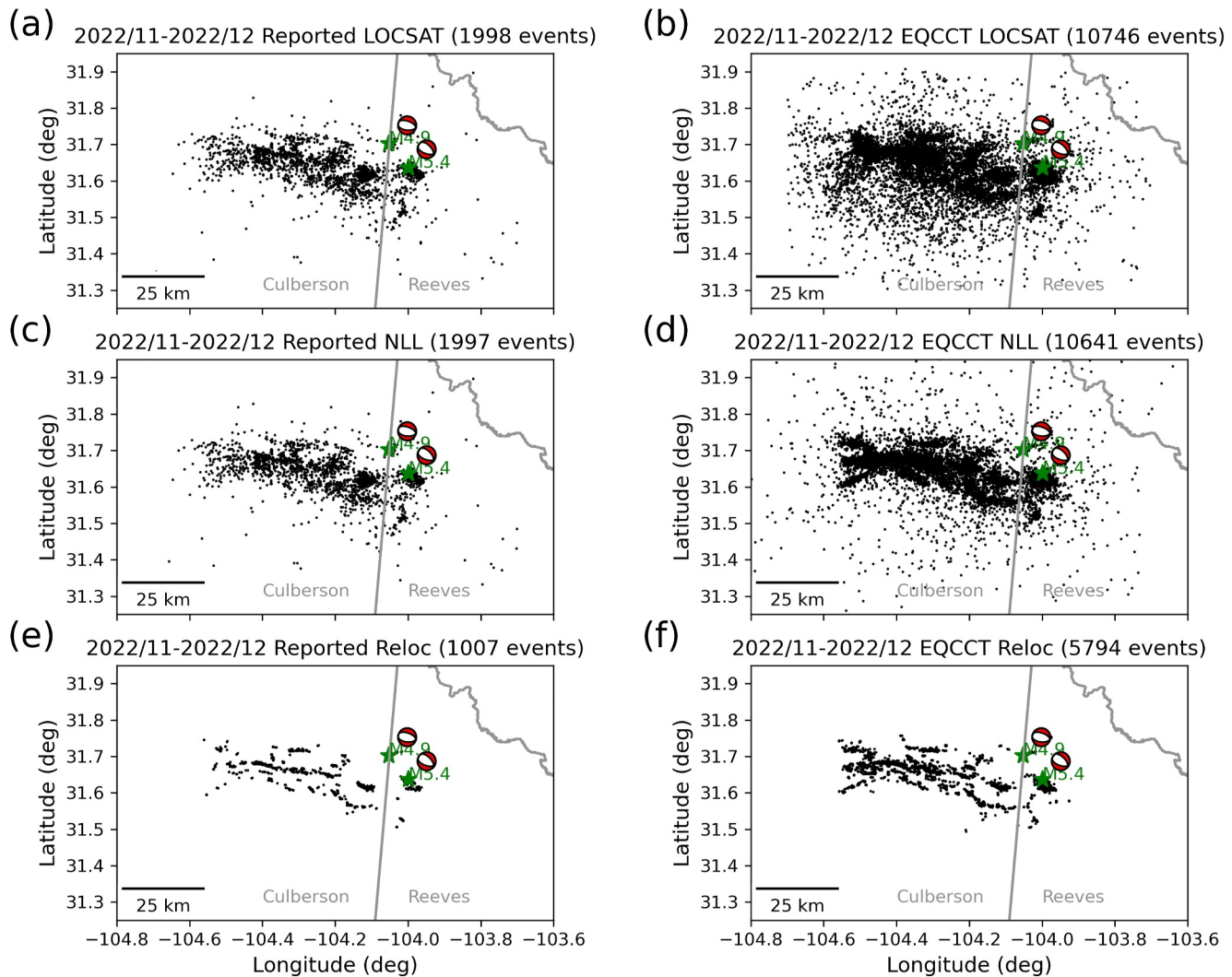


Figure 7. Comparison between the reported and EQCCT detected events along the near real-time monitoring workflow. (a) 1,998 reported events using the iasp91 velocity model detected by the traditional picker at Texnet. (b) 10,746 EQCCT detected events using the iasp91 velocity model. (c) (d) Refined locations of the reported events and EQCCT detected events using the regional velocity in the Delaware basin and the NonLinLoc method. (e) (f) Relocated locations of the reported events and EQCCT detected events using the Growclust method. All the catalogs shown here are generated automatically.

event is more reliable than the reported location. This example indicates that even for small-magnitude events, EQCCT is also more precise when used in near real-time monitoring.

To further demonstrate the evolution of the catalog through the near real-time detection workflow shown in Figure 1, we show a comparison of different catalogs between reported events and EQCCT detected events in Figure 7. Figures 7a and 7b show the initial catalog directly from the traditional picker and EQCCT picker using the iasp91 velocity model and the LOCSAT method (Kennett et al., 1995). On top of the maps are two focal mechanisms that correspond to the 2020 M4.9 Mentone earthquake and the 2022 M5.4 Coalson earthquake. Figures 7c and 7d show the refined catalog using the regional velocity model in the Delaware basin and the NonLinLoc method. The refined location process removes a few events that do not meet the location quality and makes the refined locations a little more focused compared with the initial catalog. Figures 7e and 7f show the relocated catalog after applying the Growclust method (Trugman & Shearer, 2017). It is worth noting that GrowClust is not run in real-time. It is run separately as soon as a catalog has been constructed within a certain period of time. Ideally, the GrowClust calculation can be run on an hourly basis by including newly detected events in the past hour. The event locations become more focused and well delineate the large-scale fault system in West Texas. All the catalogs presented in Figure 7 are generated automatically, which shows the significant

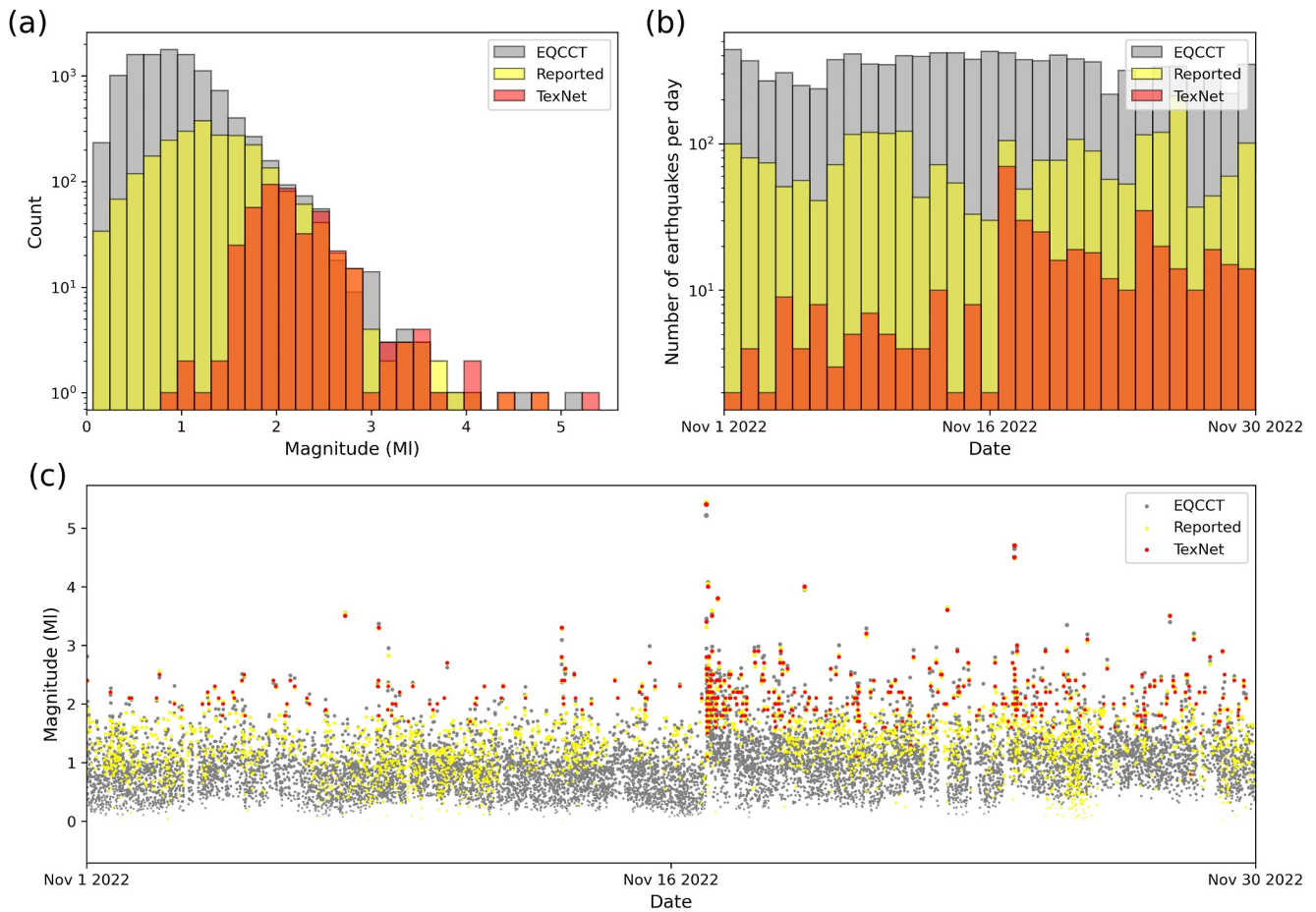


Figure 8. Event distribution for different catalogs. (a) Frequency-magnitude distribution. (b) Distribution of daily earthquakes. (c) Daily earthquake magnitude distribution. It is clear that EQCCT significantly boosts the detectability of small earthquakes daily.

advantage of the EQCCT picker in creating a highly complete catalog that facilitates regional seismotectonic analysis. Figure 8 shows event distribution in different scenarios. Figure 8a shows the frequency magnitude plot. Figure 8b shows the daily earthquake number distribution. Figure 8c shows the daily magnitude distribution, where we can clearly see that EQCCT significantly boosts the detectability of low-magnitude earthquakes.

4. Discussion

4.1. Influence of Station Density

Figure 9 shows how the station density will affect the number of detected earthquakes. Figure 9a shows the 11,563 detected events using 21 stations between 2022/10/01–2022/11/01 in the West Texas area. The station locations are plotted as inverted triangles, colored by their deployment time. As we decrease the number of stations according to their deployment time, we find that the detectability of the proposed EQCCT workflow decreases correspondingly. However, even when we decrease the number of stations to 60% of the existing stations, we still detect a huge number of events (7,533). When we further decrease the number of stations to 8 stations, the detected number of events further decreases to 4,866 events. This test also indicates that the proposed workflow is robust even in a moderately dense regional network.

4.2. Uncertainty Analysis

The uncertainty of picking models is crucial for estimating the location and relocation uncertainty. To evaluate the uncertainty of the EQCCT picks, we apply a Monte Carlo dropout Gal and Ghahramani (2016) strategy, where the model predicts the output probabilities of the arrival times several times and calculates the average. We use the

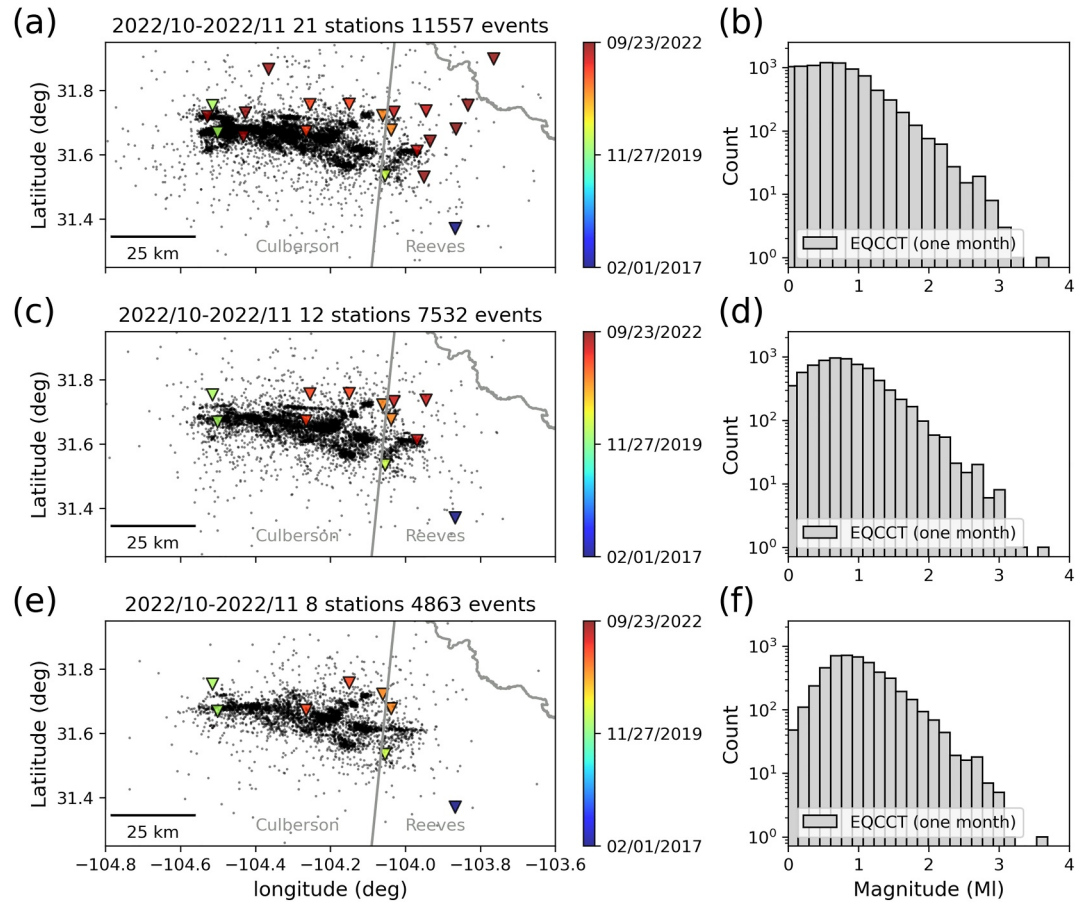


Figure 9. Station effect on the EQCCT catalog. The stations are colored according to their installation time. (a) Detected events assuming the up-to-date station distribution. (b) Event magnitude distribution corresponding to (a). (c) Detected events assuming only 12 stations exist. (d) Event magnitude distribution corresponding to (c). (e) Detected events assuming only 8 stations exist. (f) Event magnitude distribution corresponding to (e).

Monte Carlo dropout layers in the Conv and MLP blocks. We ran the EQCCT 10 times to predict the STEAD test set. Then, we obtain the standard deviation (uncertainty) of the three predictions for each test waveform. The *P* and *S*-wave uncertainty of the STEAD set is shown in Figure 10. Most of the *P* and *S*-wave picks have small uncertainty, which indicates the robustness of the EQCCT's picking performance. For the STEAD set, the average of the *P*-wave uncertainty and *S*-wave uncertainty is 0.01 and 0.03 s, respectively.

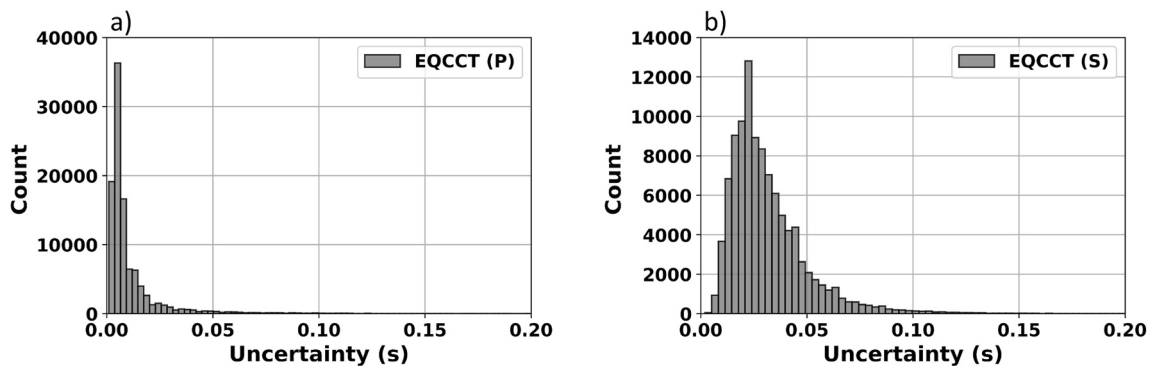


Figure 10. The uncertainty of the EQCCT for (a) *P*-wave and (b) *S*-wave picks.

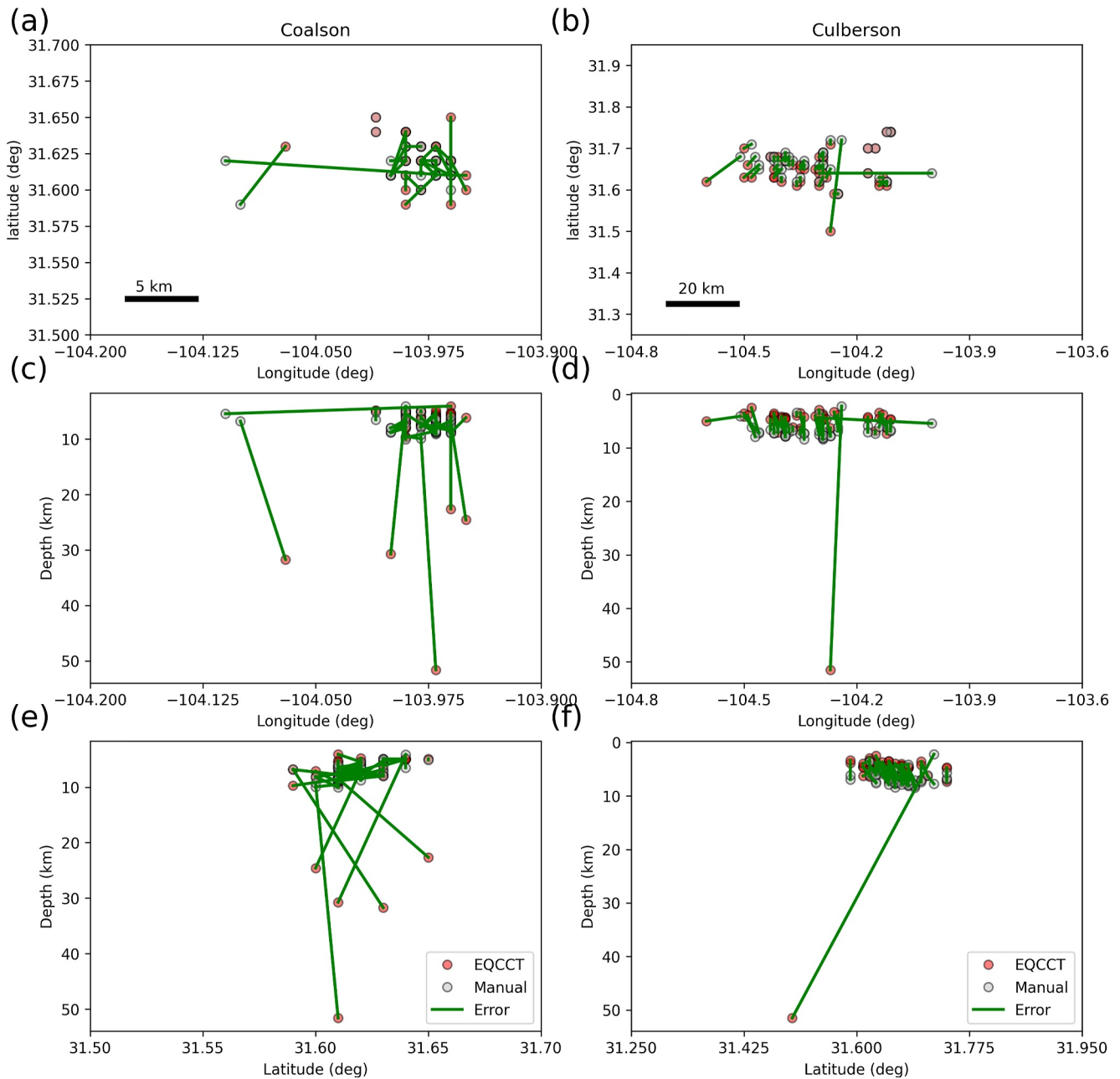


Figure 11. Comparison between EQCCT and manual locations for M2 events. Left column: Coalson area; Right column: Culberson area. The location performance is generally robust except for only several outliers in a 1-month period. (a, c, and e) Show different spatial slices for comparing EQCCT and manual locations for the Coalson area. (b, d, and f) Show different spatial slices for comparing EQCCT and manual locations for the Culberson area.

4.3. Location Outliers

Despite the fairly accurate magnitude estimation, the near real-time location using EQCCT is not 100% accurate. There are several inevitable outlier hypocenters using the current EQCCT picking model. Figure 11 shows a comparison between the automatically generated hypocenter location using EQCCT and the manually analyzed location for the aforementioned Coalson and Culberson monitoring examples. Here we only analyze the M2 events because we only manually pick the events for magnitude-above-two events. The left column (a,c, e) shows the location comparisons in longitude-latitude, longitude-depth, and latitude-depth slices for the Coalson area. Red circles denote the EQCCT location; gray circles denote the manual location, and green lines connecting each pair of locations denote the location error. The right column (b,d,f) shows the same location

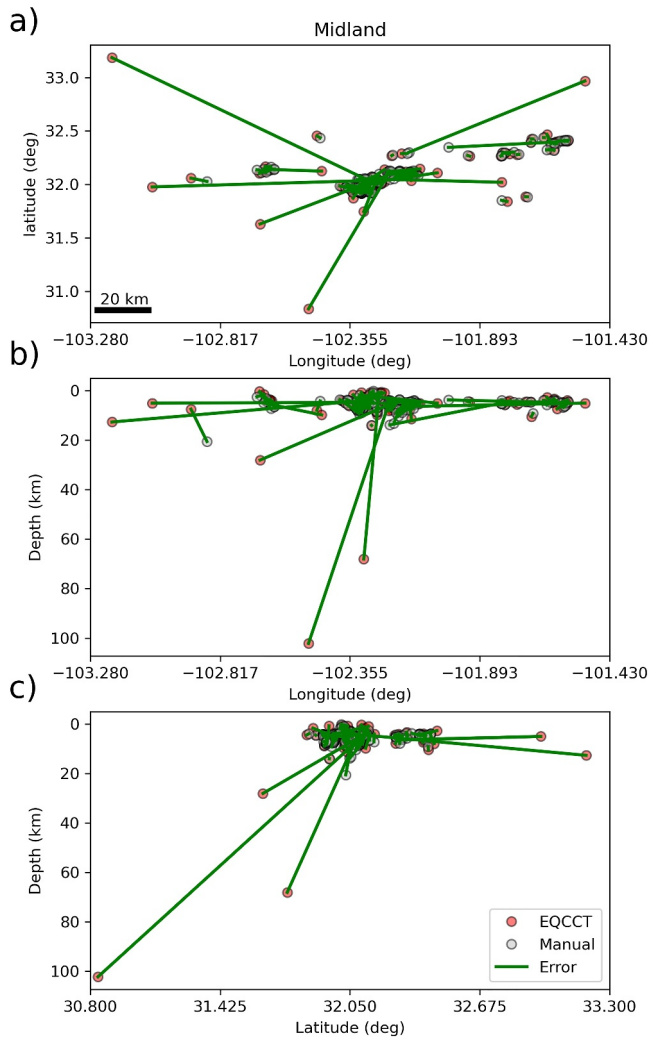


Figure 12. Comparison between EQCCT and manual locations for Midland events. The EQCCT pickers are applied to catalog events from the Midland basin. The outliers are caused by picking errors. (a–c) How different spatial slices for comparing EQCCT and manual locations for the Midland area.

comparisons but for the Culberson area. The mean absolute error (MAE) of location in the Coalson area is 2.04 km in distance and 1.34 km in depth. The MAE of location in the Culberson area is 2.59 km in distance and 2.26 km in depth. The corresponding MAEs of magnitude estimation are 0.15 MI in Coalson and 0.114 MI in Culberson. In the 1-month period in the Coalson area, there are five depth outliers (e.g., with ultra-deep depth >10 km) out of 1,471 events, resulting in an outlier rate of 0.3%. There is only one epicenter outlier (e.g., epicenter error >10 km). In the Culberson area, there is only one depth outlier out of 3,350 events, resulting in an outlier rate of 0.03%. There are five epicenter outliers with an epicenter error of about 20 km. In general, the EQCCT picker helps obtain robust and reliable near real-time location performance with a very low outlier rate due to the highly precise *P* and *S*-wave arrival picking. Possibly, location outliers can also be an association problem rather than solely a false pick problem. It is common for scautoloc associates to pick origins that should not have been associated and create origins that are very deep. To verify it, we analyze the event with a depth outlier in Figure 11f. The eventID is texnet2023caxn. Because of the abnormal depth estimation, the magnitude is wrongly estimated to be 2.2 instead of 0.54 (preferred magnitude). In this case, because the actual magnitude is small, the picks are relatively weak, causing a challenge for picking and association. Any wrongly associated pick would make the location an outlier. Not surprisingly, we check the association root-mean-squares (RMS) to be 0.879, the largest among all the events. With most associations below 0.3 for large EQCCT magnitudes, this abnormal RMS value indicates a low-quality association.

To verify the picking errors, instead of association, on location outliers, we apply the EQCCT picker to catalog data from the Midland basin and show the location results in Figure 12. In this test, the waveforms are extracted according to the catalog events (200) and manual picks. We apply EQCCT only to pick the associated events so that we can attribute the location outliers only to the picking errors. We can see that we have a slightly higher outlier rate due to the relatively sparser station distribution in the Midland basin. From this test, we understand that there is still space for further improving the deep-learning-based picking method. We also suggest the proposed workflow can be applied beyond TexNet as long as the network has a fairly dense station geometry.

4.4. Efficiency in Earthquake Analysis

Analysts' workloads have been dramatically relieved due to the proposed AI-assisted automatic detection and location workflow. On the one hand, the average speed to analyze and adjust the picks that are initially from the automatic pickers has been increased to at least three times. Traditional STA/LTA-based workflow requires analysts to adjust almost every pick on each waveform, while the new EQCCT picker makes most picks good enough, with less need to adjust the pick positions. This makes analysts' life easier on a daily basis. On the other hand, to help analysts better understand the patterns of certain earthquake swarms or aftershocks, the proposed workflow significantly increases the overall efficiency. For example, traditionally, the TexNet crew, including around 5–8 analysts, are able to analyze around 200 M2 earthquakes per month. However, due to the newly developed workflow, in half a day, we are able to analyze more than 50 times more earthquakes with the same length of data (e.g., 1 month).

5. Conclusion

Real-time monitoring requires the accurate detection, rapid location, and magnitude estimation of moderate-to-large earthquakes. The deep-learning-based EQCCT picker has been demonstrated to be robust in the West Texas monitoring areas, including the Coalson and Culberson regions. The successful applications in these areas are

attributed to the relatively higher station density. The Midland basin, however, does not enjoy the same robustness as the EQCCT picker. We have analyzed the detection, location, and magnitude estimation accuracies using the EQCCT picker for thousands of M2 earthquakes, as compared to the TexNet catalog that is from manual quality control. Most importantly, the detection recall of M2 events using EQCCT is almost 100%, making it reliable to use AI for near real-time monitoring. The tiny difference in the location and magnitude estimation errors comes from the slight picking errors in several stations where the signal-to-noise ratio is low. The robust near real-time monitoring performance enabled by the EQCCT picker suggests that we can rely on AI techniques to heavily relieve the workload of analysts, thereby enabling the analysis of smaller earthquakes.

Data Availability Statement

The data and source codes of this paper are fully open-access. The data can be downloaded via the IRS DMG. The EQCCT picker can be accessed through Saad (2024) <https://figshare.com/articles/code/Models/21865440>. The near real-time automatic earthquake monitoring code can be found at Chen (2024) <https://github.com/cheny1990/sc3flow>.

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